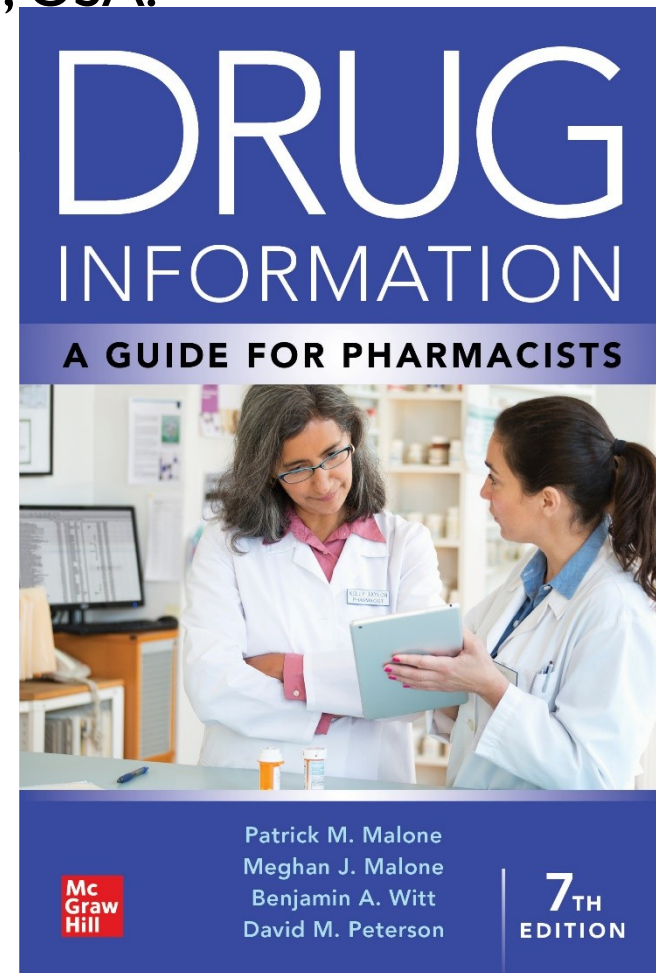


References

- I. Malone PM et al. Drug Information: A Guide for pharmacists, 6th edition, McGraw-Hill Companies, Inc., 2018, USA.



Requirements to Establish DIS

- To ensure optimal operational performance and quality provision of drug information service,
 -  appropriate personnel,
 -  adequate space,
 -  equipment and supplies .
- These resources must be located in areas that facilitate the provision of services to healthcare providers and patients/community.

Requirements to Establish DIS

- Personnel

- ✍ The number of personnel required will depend on the range of activities offered and the hours of service. The health facility should assign at least **one full-time** pharmacist.
- ✍ The drug information pharmacist(s) should have competent knowledge in **clinical use of medicines**.
- ✍ Furthermore, they shall acquire basic skills in **communication, data storage and retrieval, literature evaluation, bulletin writing, computer usage, information sourcing, clinical interventions, and documentation and report writing**.

Requirements to Establish DIS

Facilities and Equipment

- Basic facilities required for a DIS include
 - Dedicated and accessible room;
 - Furniture – desks, chairs, shelves;
 - Communications – telephone, internet access;
 - Computer, printer and accessories
 - SOPs Manual and DI formats
- Availability of other supplies such as photocopier, color printer and fax machine facilitates the DIS activities.
- A dedicated area must be set aside for the DIS that is accessible to the users.
- Allocated space should be sufficient for reference collection, storage and provision of the service.

Summary

- All pharmacists must be effective drug information providers regardless of their practice site.
- It is one of the most fundamental responsibilities of a pharmacist.
- Developing the skills of an effective drug information provider is the foundation for the pharmacist to be a lifelong learner and problem solver.

Systematic Approach to Receiving and Answering DI Questions

Chapter 2:

Learning Objectives

- *After completing this chapter, the you will be able to*
 -  Develop strategies to overcome the impediments that prevent health care professionals from providing effective responses and recommendations.
 -  Outline the steps that are necessary to identify the actual drug information needs of the requestor.
 -  Demonstrate the systematic approach to answer DI queries
 -  Formulate responses and recommendations
 -  Demonstrate DI request and response formats

Introduction

- DI pharmacists are expected to provide accurate, complete and timely response to the enquiries coming from
 - ✓ health care providers,
 - ✓ clients/patients and
 - ✓ the general public
- Therefore, a systematic approach that enables the capture of pertinent background data, and outlines a sequential method of elucidating the query, is of utmost importance

Step-wise Approach

- The following step-wise approach is recommended to respond drug information queries.

Step 1: Secure demographics of requestor

Step 2: Obtain background information

Step 3: Determine and categorize the ultimate question

Step 4: Develop strategy and conduct search

Step 5: Perform evaluation, analysis, and synthesis

Step 6: Formulate and provide response

Step 7: Conduct follow-up and documentation

Step I: Secure Demographics of Requestor

- The requestor's "profession" (e. g. , physician, pharmacist, nurse, lay person)
- The requester's name, age, location, affiliation (institution or practice if he/she is health care provider) and frame of reference (title, profession or occupation and rank)
- Mechanism for delivery of the response: telephone number(s), fax number, and/or address (mail or e-mail) or location

Step 2: Obtain Background Information

- General questions for obtaining background information
 - ✍ Whether the request is **patient specific** or **academic**
 - ✍ The resources that the requestor already consulted
 - ✍ If the query is patient specific
 - ✓ Information regarding diagnosis
 - ✓ Specific patient laboratory values
 - ✓ Other medications, and pertinent medical information
 - ✍ The urgency of the request (negotiate the time)

Step 2: Obtain Background Information

- Why background information?
 - ❖ The question may not be stated concisely or the requestor may not know how to ask the question
 - ❖ To formulate an acceptable response, both the caller and researcher must have a clear understanding of the ultimate question
 - ❖ To develop a more complete picture of the question

Step 2: Obtain Background Information

Examples

- “what is the dose of amoxicillin (Amoxil®)?”
 - ✍ This question could be answered quickly (and potentially inaccurately) by stating that the normal dosage as 500 mg every eight hours
 - ✍ Should be answered by gathering background information concerning the origin of the question
 - ✍ The dose of this antibiotic depends upon a no of factors: even to the extent that amoxicillin mayn't be the best agent to select for the infection₁₃

Step 2: Obtain Background Information

Examples

- **Patient:** Is Dextromethorphan good for cough?"
- **Pharmacist:** "Yes, Dextromethorphan is one of several effective medications for cough. How much have you been taking?"
- **Patient:** "I've only been taking two teaspoonfuls in the morning. Should I take more?"
- **Pharmacist:** "Do you have fever?"
- **Patient:** "No, not that I know of."
- **Pharmacist:** "Do you have a productive cough or chest pain when you breathe?"
- **Patient:** "No."
- **Pharmacist:** "Well, it's probably okay to take two teaspoonfuls every 6 hours as needed for cough. If the cough doesn't go away in the next week or so, you should contact your doctor."

Step 2: Obtain Background Information

Examples

Patient: "Can you check a drug interaction? I've been on so many medicines. The doctors don't know what to do with me anymore."

Pharmacist: "Your doctor did not know what to do with you because you have been on several medications and did not work?"

Patient: "Yes, I think he does not want to see me any more."

Pharmacist: "Can you tell me what medicines you have tried in the past?"

Patient: "I have high blood pressure. I've tried everything, and everything I have ever taken has made me feel bad,"

Pharmacist: "If I understand you, Mr. T, you are telling me that you believe any new HTN medicine is going to make you feel bad."

Patient: "That's what I believe. The doctors told me that enalapril ruined my kidneys. The Esidrix that I used before didn't make me feel bad, but it didn't do any good. This nifedipine is really making me dizzy, and I can hardly stand up."

Step 2: Obtain Background Information

- What do we learn from these cases?
 - (1) People do not know what the problem is & they tend to generalize
 - (2) It is the pharmacist's job to help find the problem

Annex 3. I: Drug information request form

ABC Facility- Drug Information Service

DRUG INFORMATION QUERY FORM									
				Date:					
				Time:		am/pm		am	
I-Requester's contact information: (To be filled by requester or DI pharmacist)									
Full Name:		Dr. Mr. Mrs. Sr.):		Mr					
Physical address:		Tel No:				E-mail:		Fax:	
Qualification/Profession:		GP		Specialist		Health Officer		Patient	
		Nurse		Pharmacist		Druggist		Student	
		Other (Please specify)							
Method of Contact:		Walk-in		Phone		Written form		E-mail	
		Fax		Letter		Other (Please specify)			
II-Background information on query:									
The request is:		Patient specific		Academic		Other			
If Patient specific, Please record patient information that you feel may be helpful in answering your request (such as patient's age, sex, weight, disease states, laboratory values, medications allergies etc...).		Age		Sex		wt (kgs)		Diagnosis	
		Current Medication:							
		Concurrent medication/s:							
		Allergies:							
		Other information related to Patient:							
Request/Question:									
Preferred method of Response:		Verbal		Phone		printout		E-mail	
		written form				Fax			
Response needed in:		prompt		30-60 min		end of day		when time permits	
References required:		YES		NO					
Additional Information required:									
Initials of the requester(optional)				Date:				Time:	
For DI pharmacist use only									
Enquiry Received on (Date):				Time:					
Enquiry Received by:				Enquiry Reference No.		DI001			
Response given to requester on (Date):				Time:					
Response made by:									

Step 3: Determine and categorize the ultimate question

- ✍ The most critical one in establishing a good search strategy
- ✍ Depends largely on the adequacy of the background information obtained
- ✍ The first step is putting the pieces of information together to form the ultimate question
- ✍ Some take the ultimate question as the final iteration of the question
- ✍ Once the ultimate question has been determined, the next step is to categorize the question
- ✍ It is very important that the requestor confirms the ultimate question prior to categorization & the development of a search strategy

Step 3: Determine and categorize the ultimate question

Develop a time line for response

- Completely understanding the scope of the "true" question also aids in developing a realistic estimate of the time required to compose a response

Categorize the question: Why??

- Allows for efficient use of resources by providing the foundation of a logical progression process
- An all-inclusive resource with data to answer every drug information question does not exist
- References contain specific types of information
 - ✓ Numerous topic specific resources are available (e.g., drug interactions, infectious disease, ADRs, lactation & pregnancy...)

Step 3: Determine and categorize the ultimate question

- Classification of a request aids in developing a more effective search strategy
- Selecting the resource with the highest probability of containing the desired information can decrease the time requirement and increase the accuracy of the response
- Otherwise, unnecessary time and energy may be expended on searching references unable to produce the needed facts
- DIC should have lists of standard categories that they use

Step 3: Determine and categorize the ultimate question

Examples of drug information categories

☒ Adverse Effects	☒ Identification of a product
☒ Availability	☒ Pharmacokinetics
☒ Compatibility/Stability	☒ Pharmacology
☒ Compounding	☒ Poisoning/Toxicology
☒ Dosing and Administration	☒ Pregnancy and Lactation
☒ Drug Interaction	☒ Therapeutic Use

Examples

 **Physician:** “What is the dose of ceftazidime?”

 **Pharmacist:** "Is this question about a patient?"

 **Physician.** "Yes."

 **Pharmacist:** "Why do you need to give ceftazidime to this patient?"

 **Physician:** "Well, the patient has most likely developed a cellulitis. Culture samples were taken from the infection site. However, we would like to initiate an antibiotic therapy empirically."

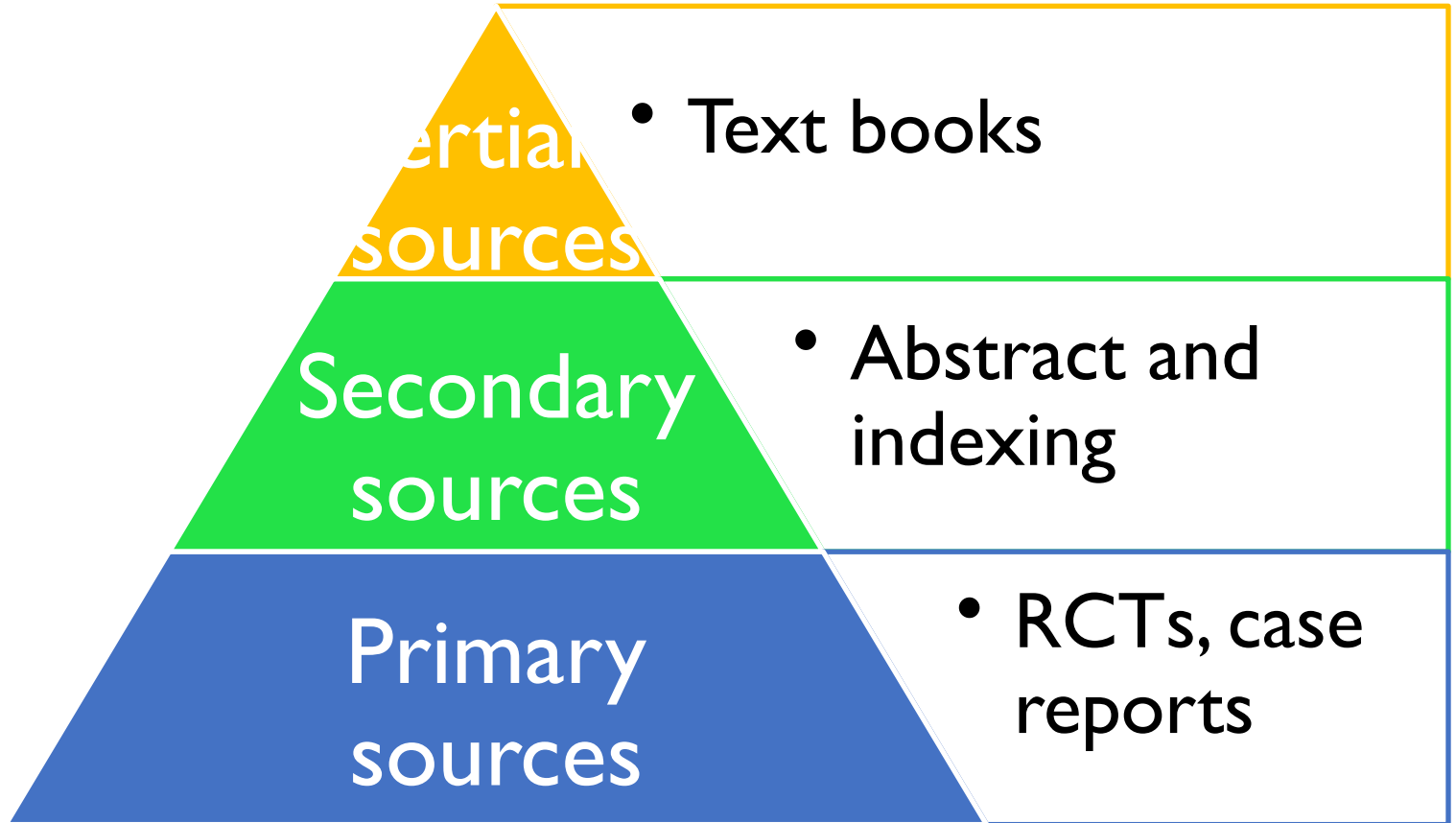
Examples

- Create your own scenario by asking and generating background information response for the following requests.
 - ☑ Physician: What is the dose of **ceftriaxone**?
 - ☑ Physician: What is the best alternative drug to use in place of **cloxacillin**?
 - ☑ Physician: Do we have **enoxaparin** in this hospital?

Step 4: Develop strategy and conduct search

- Prioritizing resources based on their probability of containing the desired data or information will help in pin-pointing the response.
- The desired information is then collected from each of the sources.
 - ✓ e.g., if question categorized as “adverse effect”, the adverse effect oriented resources will be used
- The search process involves a search that starts with the tertiary literature, followed by the secondary literature and then to the primary resources.

Step 4: Develop strategy and conduct search



Step 4: Develop strategy and conduct search

- Sometimes a search will require all 3 levels of the pyramid
 - ☑ A tertiary source to familiarize yourself with the topic
 - ☑ Secondary to identify appropriate primary literature
 - ☑ Carefully evaluated primary literature
- Quite often tertiary source will be sufficient for:
 - ☑ Information for a patient consultation
 - ☑ Availability of dosage strengths or forms or compatibility information
 - ☑ Active ingredients of preparations
 - ☑ EFDA approvals, US FDA approved or UK licensed

Step 4: Develop strategy and conduct search

- When you get a drug information question where do you start? Do you go to the **internet** or to the **Library**, or do you start with **NEJM**?
- Point of entry for searching:
 - ✍ Starting point for searching
 - ✍ The special term for the place you start searching is called: point of entry

Step 5: Perform evaluation, analysis, and synthesis

- DI provider should take time to evaluate the information, analyze it, and then synthesize it into a good reply
- **Evaluation:** Evaluating the quality of the information is key to a good response
- **Analysis:** of the found information must be done with the requestor demographics in mind and the category of the question
- **Synthesis:** involves the careful integration of critical information about the patient, disease, and medication along with pertinent background information to arrive at a judgment or conclusion.

Step 6: Formulate and provide response

- Formulated response must be to the point, yet adequately comprehensive.
- Response to a question must be timely, current, accurate, complete, concise, well referenced, clear and logical, objective and balanced, free of bias or flaws, applicable to specific circumstances, answers important related questions and addresses specific management of patients or situations.
- The requester might choose getting the response verbally, via telephone or over a physical conversation

Annex 3. 2: Drug information response form

ABC Facility- Drug Information Service

DRUG INFORMATION RESPONSE FORM									
					Enquiry Reference No.:				
					Date:		Time		
To(name of inquirer):				Phone No:		Email:			
Dear									
We acknowledge the receipt of your enquiry on drug information dated									
under ref. No									
We are pleased to put forward the required information as follows:									
Question/query:									
Answer/response:									
References:									
Additional information/materials and recommendations provided:									
Disclaimer:									
The DIS is designed to assist health care providers and other users to provide accurate, up-to-date, reliable and complete									
We hope we have served you with this information and in case you need further information/materials, please fill free to									
Response completed By:				Date:		Initials:			

Step 7: Conduct follow-up, documentation and reporting

- Follow-up is the process of verifying the appropriateness, correctness, and completeness of a response following the communication
- Follow-up allows you to know if your recommendations are accepted and promptly implemented
- Follow-up allows pharmacists to receive valuable feedback from requesters and to learn from the experience
- Documentation is essential for reducing liability and potentially promoting the development of a continual service

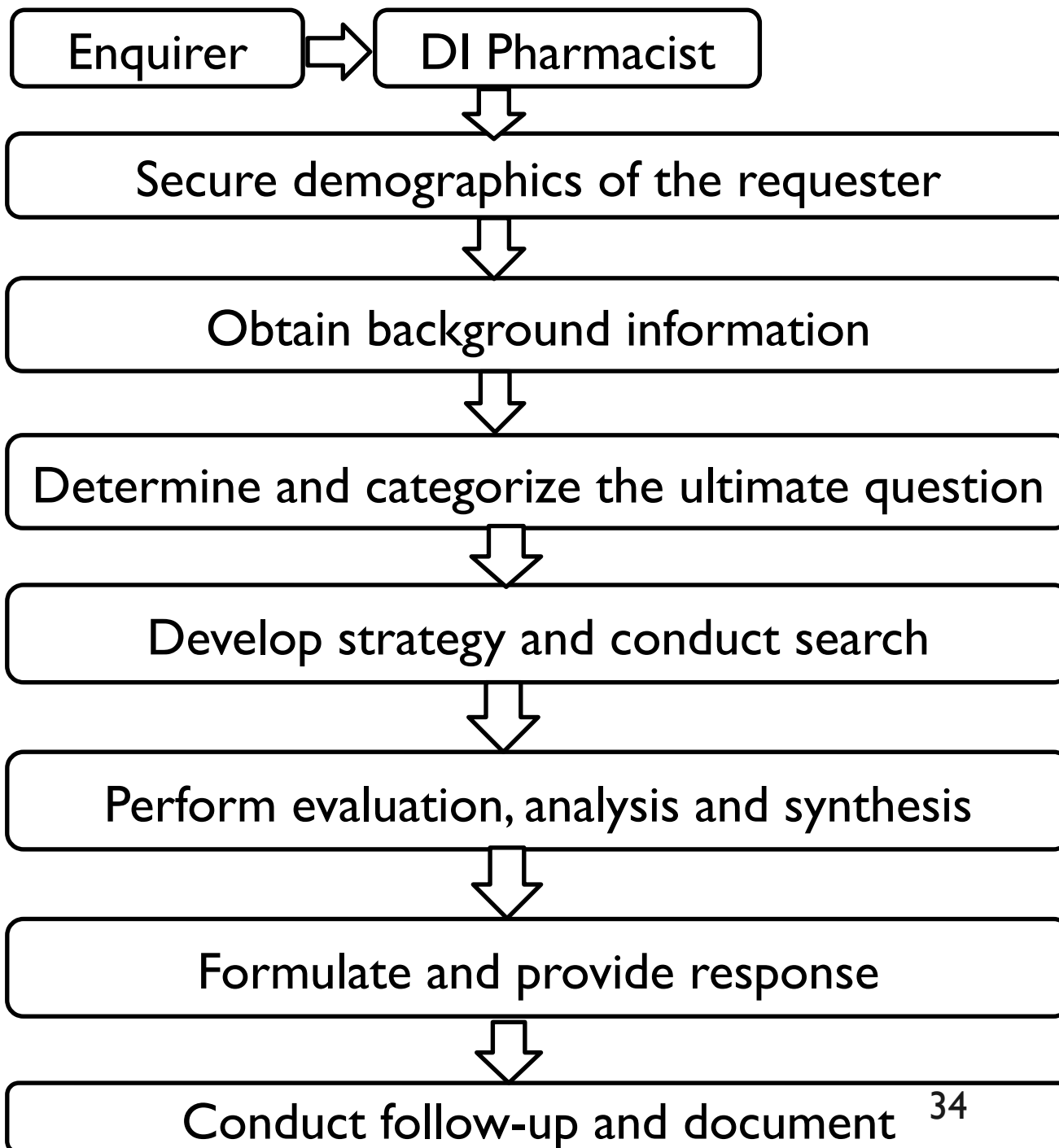
Step 7: Conduct follow-up, documentation and reporting

- **Follow through**

- ✍ is the process of readdressing a request based on the availability of new data or a change in the situation or circumstances that were decisive in the formulation of the first response
- ✍ For example, the basis of a decision to use a novel therapy in a patient may be confirmed or refuted according to a new article
- ✍ In the same scenario, the development of renal failure (as a co-morbidity) in this patient may prompt an update of the original response

Annex 3. 3: Drug information service feedback form

<u>DRUG INFORMATION SERVICE FEEDBACK FORM</u>									
					Enquiry Reference No.:				
					Date of enquiry:				
Dear enquirer:									
We want to hear from you									
The _____ hospital/H center DIS is seeking your feedback on the information we have provided in response to your enquiry under _____ Dated _____ . We value your feedback because this helps us to stay in touch with your needs and for the continuous quality improvements of _____									
We invite you to use this form to submit feedback or complaint. Provision of the information requested									
1. Was the information received in time?					Yes	No			
2. Was the presentation of the information					Yes	No			
3. Did the information provided meet your					Yes	No			
4. Was the information used?					Yes	No			
5. If your answer to Question No. 4 is Yes, what was the outcome?									
6. If your Answer to Question No. 4 is No, please describe the reason why the									
7. Additional comments and suggestions on the DIS									
Name of enquirer:					Phone No:		Email:		
We thank you for your time and response									



Drug Information Resources

Chapter 3

Sources of drug information

- Although basic information about drugs is obtained through training in pharmacy profession, additional knowledge can be gained from various sources.
- These sources of drug information can be classified into:
 - Primary,
 - Secondary and
 - Tertiary

Tertiary sources

- Tertiary resources provide information that has been *filtered and summarized* by the author or editor to provide a quick and easy summary of a topic.
- Some examples of tertiary resources include
 -  Textbooks,
 -  DI databases like Micromedex,
 -  Review articles in journals,
 -  Clinical guidelines, and other general information

Tertiary resources

- Some tertiary resources are available via online access as electronic books (eBooks) or electronically searchable online application
- Examples:
 -  Micromedex®,
 -  Clinical Pharmacology®,
 -  Lexicomp®.
- These resources will provide the practitioner with general information needed to familiarize the reader with the topic

Tertiary resources

- **Benefits of tertiary resources**

- ✍ Tertiary resources are convenient, easy to use, as well as being familiar to most practitioners.

- ✍ Most of the basic information needed by a practitioner can be found in these sources, making these excellent **first-line resources** when dealing with a DI question

Tertiary resources

Limitations of tertiary resources

- Many tertiary resources are available in both hard copy (print) and electronic formats (e.g., eBooks, electronically searchable DI databases).
 - ✍ The major drawback to **print copy** tertiary resources, however, is the **lag time** associated with updating information. Medical information changes very rapidly; thus, it is possible that information may be out of date before a text is even published.
 - ✍ **Electronically available** tertiary resources, such as eBooks and DI databases (e.g., Micromedex®, Lexicomp®), have helped this situation in some cases; however, the requirement for information to be reviewed and summarized requires an inherent delay in communicating new information.

Tertiary resources

- It is also possible that information in a tertiary resource may be **incomplete**, due either to **space limitations** of the resource or **incomplete literature searches** by the author
- Other problems that can be seen with tertiary information include errors in transcription, human bias, incorrect interpretation of information, or a **lack of expertise by authors**

Evaluation for tertiary literature

1. Does the author have appropriate experience/expertise to publish in this area?
2. Is the information likely to be timely, based on publication date?
3. Is information supported by appropriate citations?
4. Does the resource contain relevant information?
5. Does the resource appear free from bias and blatant errors?
6. Is the information most updated version of the resource?

Tertiary resources

- **Format of tertiary resources**

- ✍ **Electronic resources** are often preferred because:

- ✓ They may be easier to use (a keyword search and an instant return of search results),
 - ✓ Allow quicker access to information,
 - ✓ Allow multiple searches to be performed simultaneously, and
 - ✓ Often contain more recent information available regarding a topic than is found in the hard copy version of a reference.

- ✍ Additionally, many electronic networked resources allow use of the same resource at more than one location.

- ✍ This lets many practitioners access information from a variety of physical locations rather than being restricted to only medical libraries or DI centers.

- ✍ **Mobile apps** have no geographic restriction to accessing and retrieving information once downloaded.

Tertiary resources

- **Tertiary resources for mobile devices**

- ✍ The increasing incorporation of mobile devices into clinical practice settings has prompted an expanding choice of DI databases for that medium.
- ✍ Many of the major compendia available electronically also offer a product for a mobile device
- ✍ Examples: Medscape®, UpToDate®, DynaMed Plus®, EvidencePlus®, DynaMed®, and Epocrates®

Tertiary resources

General, Product-Related

- ☒ Drug Facts and Comparisons, Micromedex, prescribing information, AHFS Drug Information

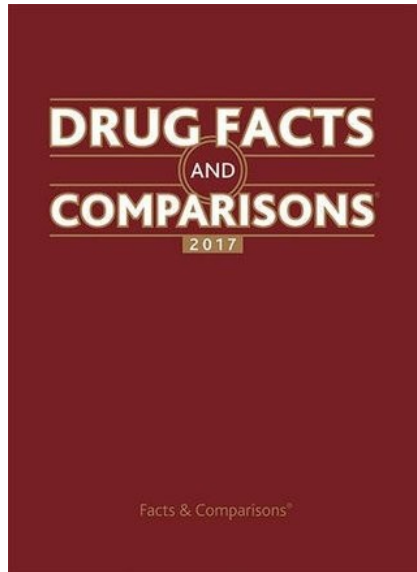
Adverse Effects

- ☒ Meyler's Side Effects of Drugs, Side Effects of Drugs Annual, prescribing information, PDR, FDAble (Medwatch)

Compatibility, Stability

- ☒ Handbook of Injectable Drugs, King Guide to Parenteral Admixtures, USP–National Formulary (NF)

Tertiary resources



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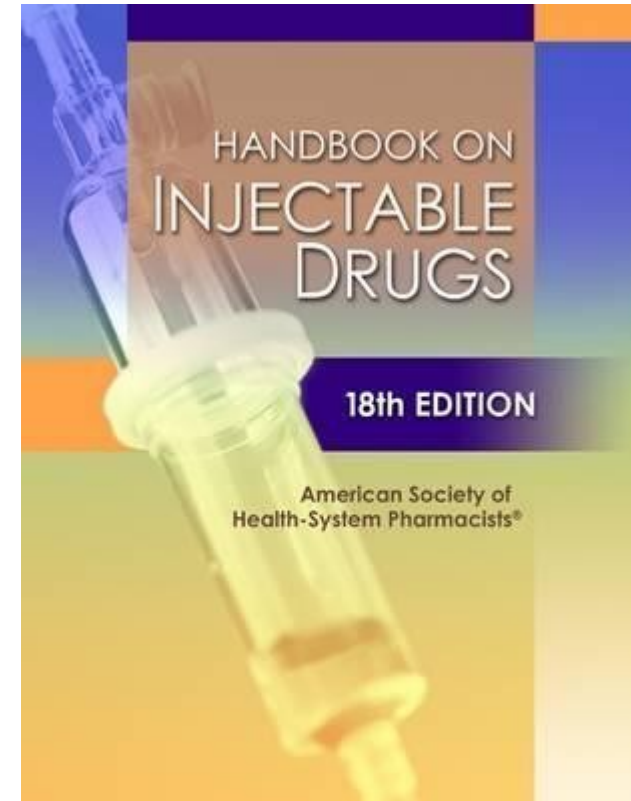
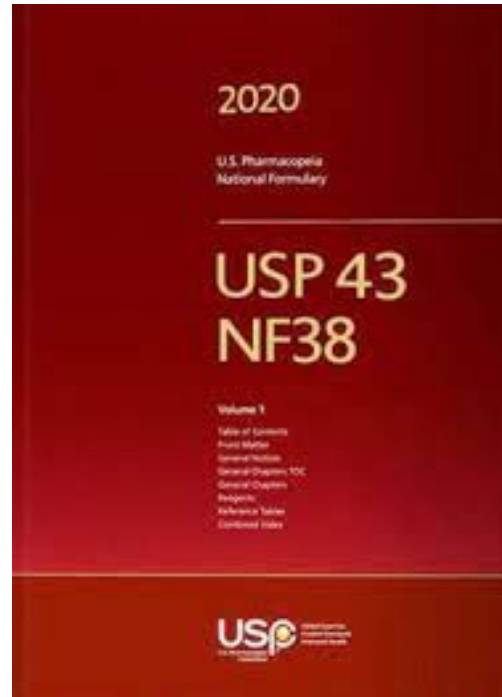
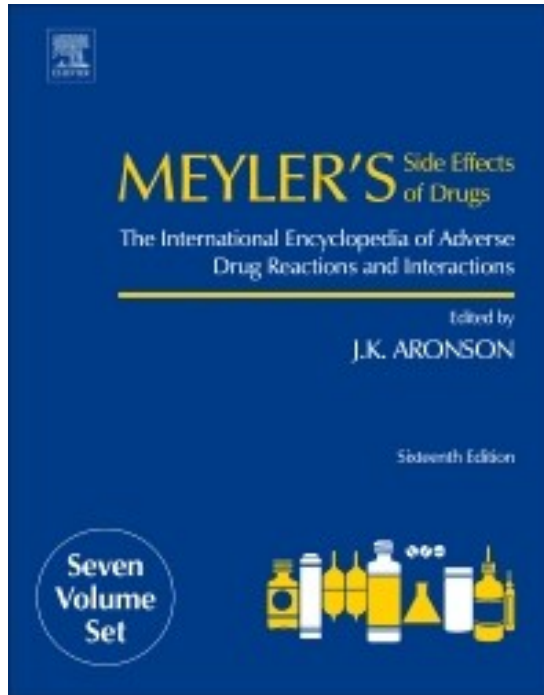
Resources

- Black Box Warnings
- Comparative Tables
- Do Not Confuse Drug List
- Drug Classes
- Drug Consults
- REMS

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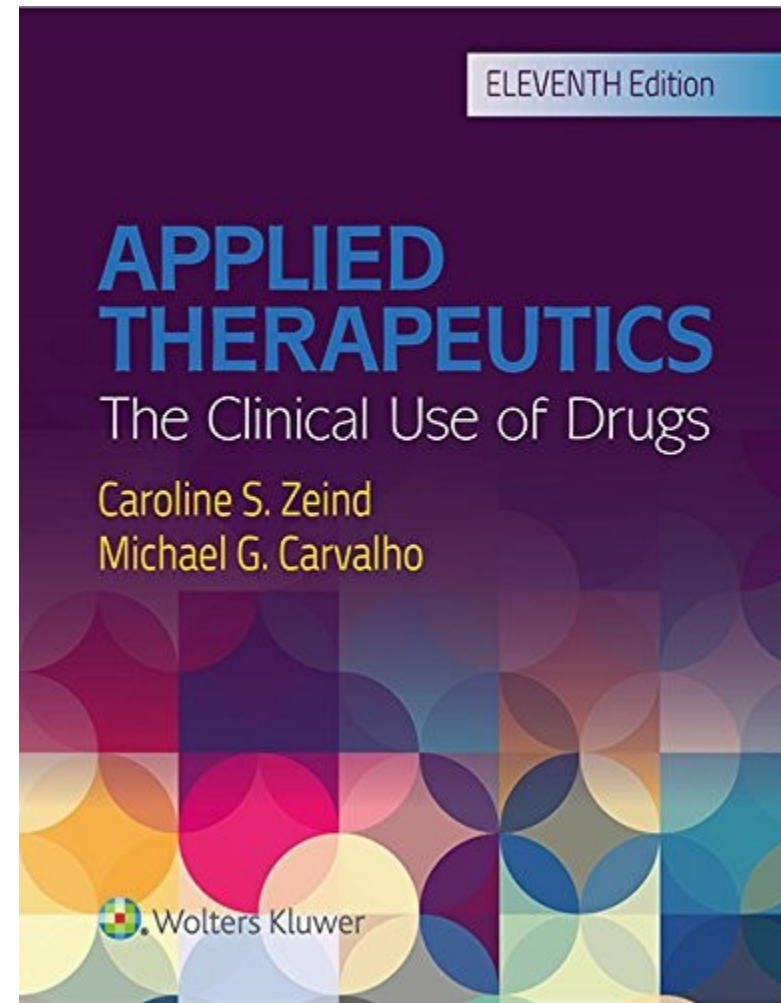
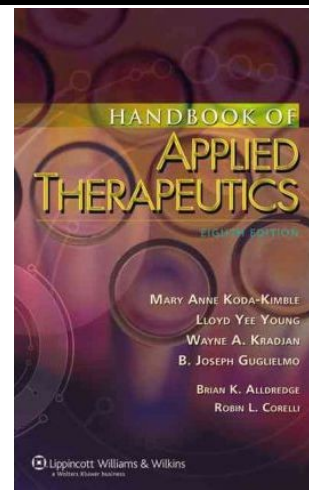
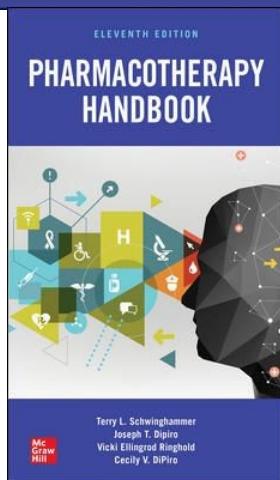
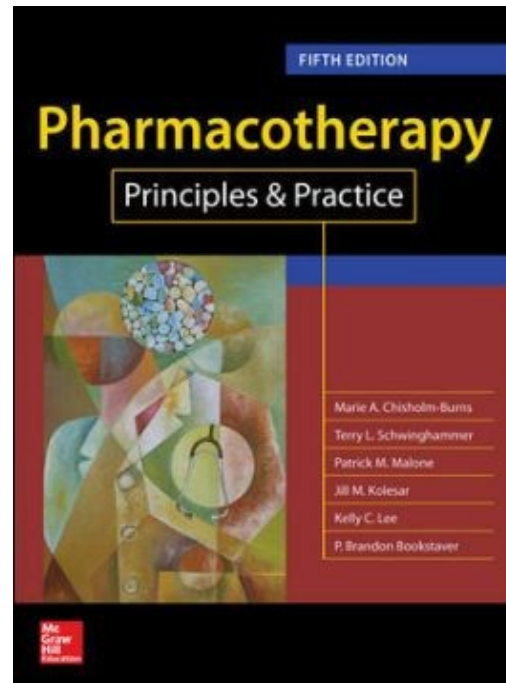
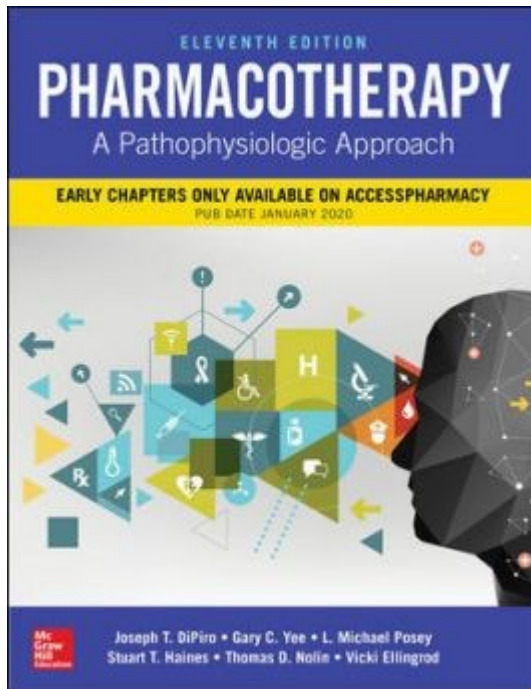
Tertiary resources



Tertiary Resources

- **Dietary Supplements, Herbal Supplements:**
 - NIH's Dietary Supplement Label Database, MedlinePlus, AltMedDex, Review of Natural Products
- **Diseases, Diagnoses:**
 - Pharmacotherapy: A Pathophysiologic Approach, Pharmacotherapy Principles and Practice, Applied Therapeutics: The Clinical Use of Drugs, The Merck Manual of Diagnosis and Therapy, Harrison's Principles of Internal Medicine
- **Drug Interaction**
 - Hansten and Horn's Drug Interaction Analysis and Management, Drug Interaction Facts, Stockley's Drug Interactions, major compendia

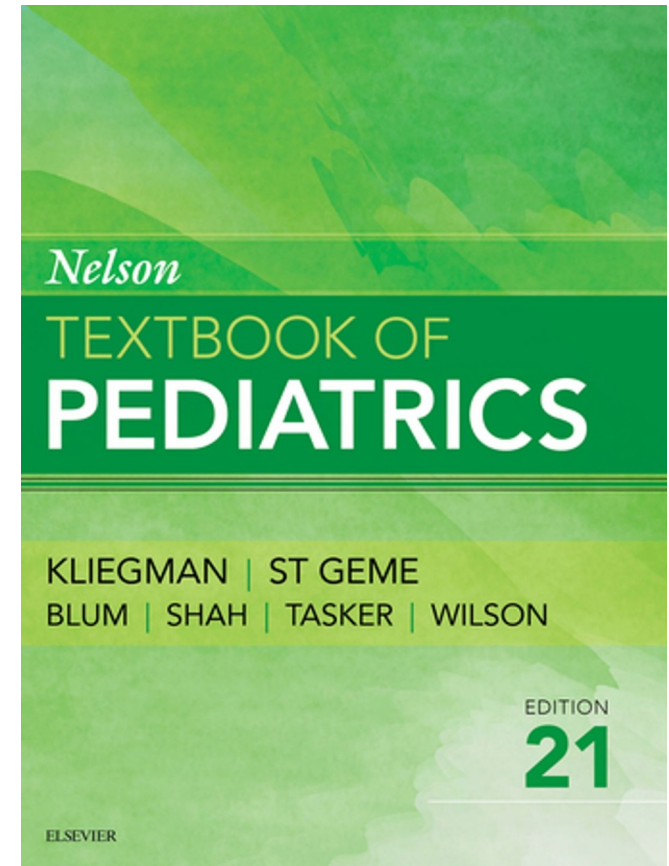
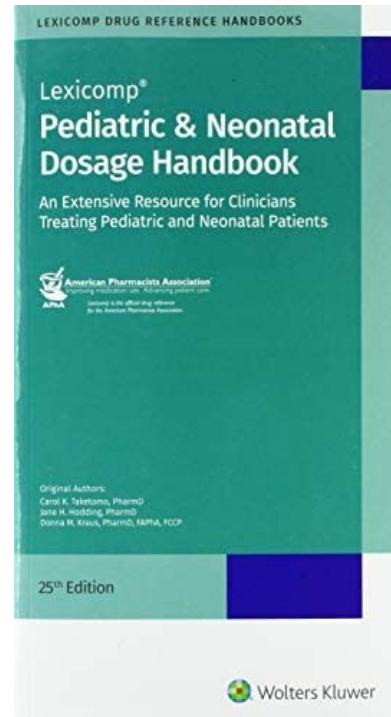
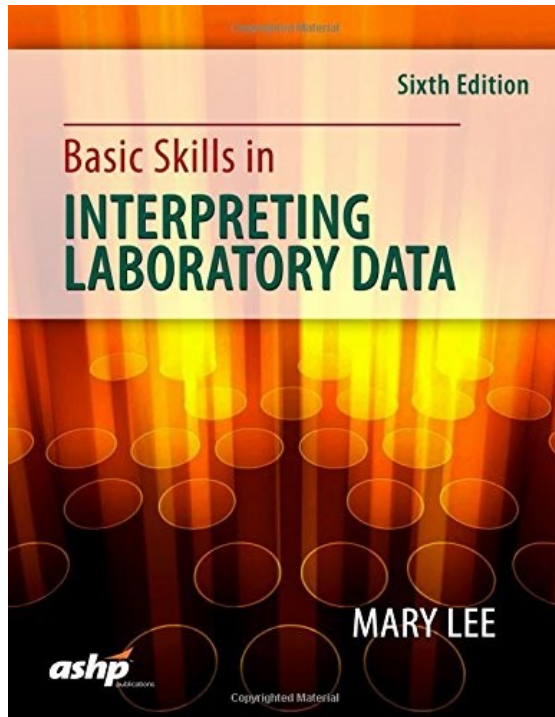
Tertiary resources



Tertiary resources

- **Drug Use in Renal Dysfunction**
 - ✍ Drug Prescribing in Renal Failure: Dosing Guidelines for Adults (Aronoff)
- **Laboratory Tests**
 - ✍ Basic Skills in Interpreting Laboratory Data (ASHP), Mosby's Manual of Diagnostics and Laboratory Test (Pagana), Pocket Guide to Diagnostic Tests (Nicoll)
- **Pediatric Pharmacotherapy**
 - ✍ Pediatric and Neonatal Dosage Handbook, Nelson Textbook of Pediatrics
- **Pharmaceutical Calculations**
 - ✍ Pharmaceutical Calculations (Zatz), Pharmaceutical Calculations (Ansel), Lexi-Calc
- **Pharmacokinetics**
 - ✍ Basic Clinical Pharmacokinetics, Applied Biopharmaceutics and Pharmacokinetics,

Tertiary resources



Tertiary resources

- **Pharmacology**

- ✍ Goodman & Gilman's: The Pharmacological Basis of Therapeutics, Basic & Clinical Pharmacology (Katzung and Trevor)

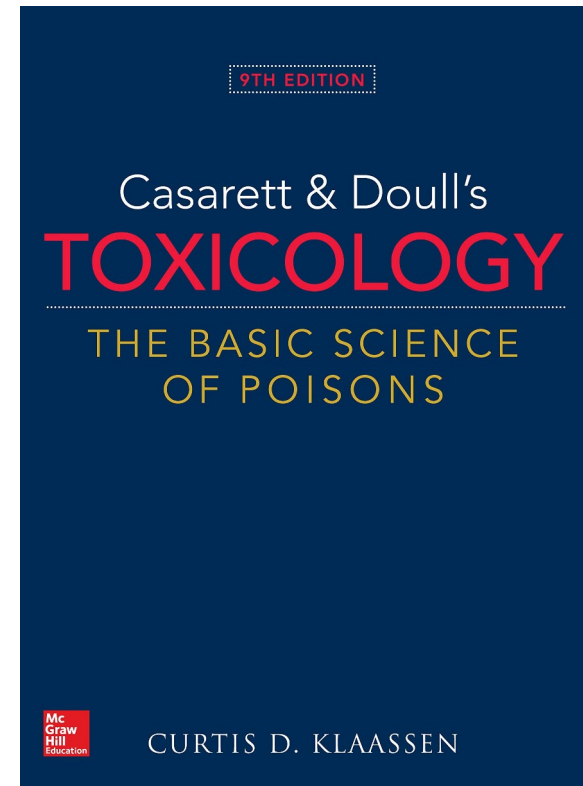
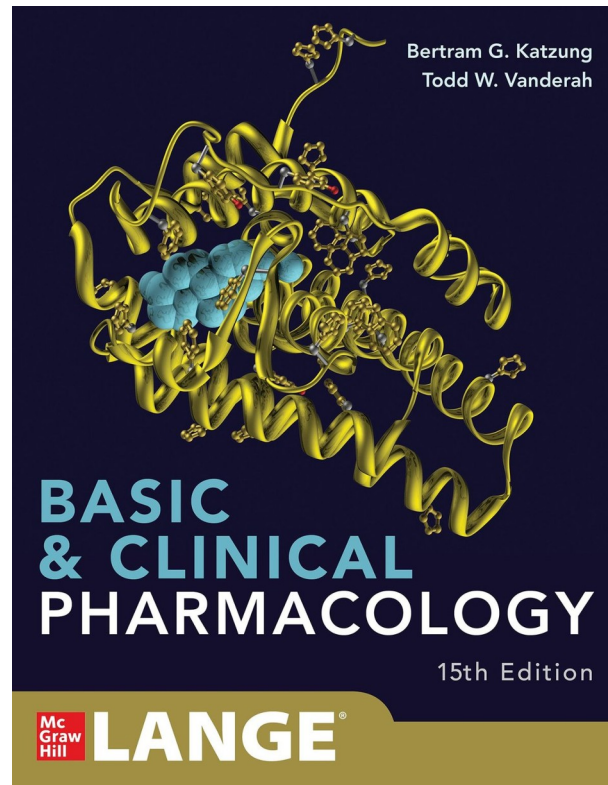
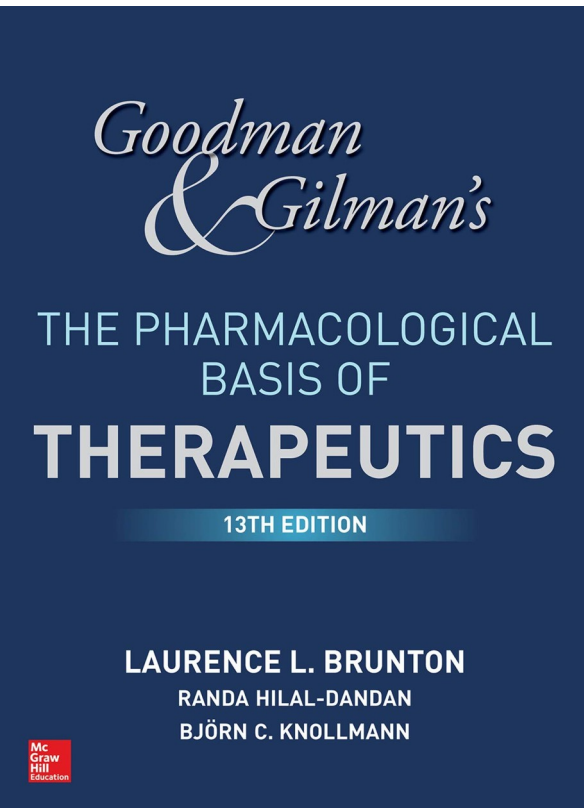
- **Pregnancy, Lactation**

- ✍ Briggs Drugs in Pregnancy and Lactation, LactMed, Medications and Mother's Milk

- **Toxicology**

- ✍ OXNET, POISINDEX, Lexi-Tox, Goldfrank's Toxicologic Emergencies, Casarett & Doull's Toxicology: The Basic Science of Poisons

Tertiary resources



Case Study

- A 15-year-old patient has recently been started on **atomoxetine** for treatment of ADHD. He is taking no other medications. He has noted recently that his hair is thinning and wants to know if this might be drug related.
 - *What are appropriate tertiary resources to consult for a response to this request?*

Case study

- A pharmacy student is working on a presentation involving illicit drugs. She knows that there have been recent news stories about adolescents using **tramadol** for recreational use, and she is curious at what dose this product is toxic.
 - ❓ *Which resources would be useful for her project?*
 - ❓ *What search terms might she utilize?*

Secondary resources

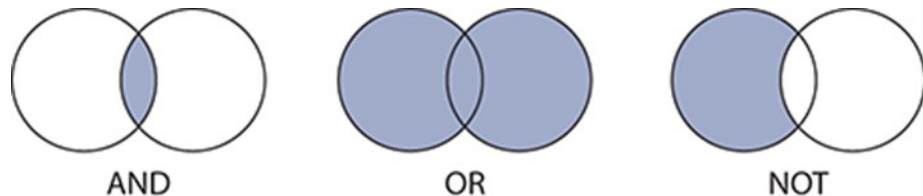
- *Secondary resources are mainly in the form of searchable **database** that enables location and retrieval of primary or tertiary resources.*
- In two forms
 -  **Indexing** consists of providing bibliographic citation information (e.g., title, author, and citation of the article),
 -  while **abstracting** also includes a brief description (or abstract) of the information provided by the article or resource cited
- This category of resources is similar to an old **cataloging** system.

Secondary resources

- There are various secondary resources available either as
 - Paid (e.g., EBSCOhost, Excerpta Medica, CINAHL) or
 - Free (e.g., PubMed, Cochrane Library)
- Using a secondary resource allows more efficient and expedient search for either primary literature (e.g., a clinical trial) or tertiary literature (e.g., a review) for a given topic.

Searching Secondary sources

- Electronic searches generally use the **Boolean operators AND, OR, and NOT**.
- The operator **AND** will combine two terms, returning only citations containing both of those concepts or terms.
- Combining two terms with the operator **OR** will result in an equal or greater number of returns since it will include any citation where either term is used.
- Use of the term **NOT** should be used with caution, as it will always decrease the number of returns.



Secondary resources

- Cochrane Library
- Cumulative Index to Nursing and Allied Health Literature (CINAHL)
- Google Scholar
- International Pharmaceutical Abstracts (IPA)
- MEDLINE®

Case Study

- A physician requests information about the use of **sildenafil** for treatment of female sexual arousal disorder. She also requests information about use of any of the other phosphodiesterase-5 inhibitors.
 - ❓ *What resources might be good places to look for this information?*
 - ❓ *What search terms should be used?*
 - ❓ *Should limits be utilized?*

Primary resources

- **Primary resources** include **clinical research studies** and reports, both published and unpublished.
- *There are several types of publications considered primary, including controlled trials, cohort studies, case series, and case reports.*
- Primary resources often provides the **most recent** and **in-depth information** about a topic, and allows the reader to analyze and critique the study methodology to determine if the conclusions are valid
- Typically, primary resources require a longer time commitment and a skilled expertise in **literature evaluation** to accurately assess the value and application to health care.

Primary resources

- **Advantages** to the use of primary literature include
 - (1) access to detailed information about a topic and the ability to personally assess the validity and applicability of study results.
 - (2) primary literature is more recent than tertiary or secondary literature.
- However, there are several **disadvantages** to the use of primary literature alone, which include
 - (1) misleading conclusions based on only one trial without the context of other research,
 - (2) the need to have good skills in evaluating the literature (e.g., clinical trial evaluation) and
 - (3) the time needed to evaluate the potentially large volume of literature available.

Common primary sources

- American Journal of Health-system Pharmacists (AJHP) <http://www.ajhp.org/>
- New England Journal of Medicine (NEJM) <http://www.nejm.org/>
- Journal of the American Medical Association (JAMA) <https://jamanetwork.com/>
- Ethiopian Pharmaceutical Journal: <https://www.ajol.info/index.php/epj>
- Ethiopian Journal of Health Development: www.ejhd.org/
- Ethiopian Medical Journal - <http://www.emaethiopia.org/>, www.emae mj.org/
- Ethiopian Journal of Health Sciences- <https://www.ejhs.ju.edu.et/>

Levels of Evidence



Alternative Sources

EFDA: <http://www.fmhaca.gov.et/>

The screenshot shows the EFDA website with a dark blue header. The header contains the EFDA logo, the name in Amharic and English, and links for English and Amharic languages. Below the header is a navigation bar with links: Home, About Us, News and Events, Services, Publications, Directorates, COVID19, Announcement, and Contact. The main content area features a large image of blue and white capsules. On the left, the 'i-Import' section is highlighted with a yellow underline. Below the title, a paragraph describes the i-Import application. A 'VIEW DETAILS' button is located below the text. At the bottom, there is a row of six small images with labels: 'Welcome to EFDA!', 'Our Vision', 'i-Import', 'i-Register', 'i-License', and 'Traceability'. A 'Privacy - Terms' link is visible in the bottom right corner.

i-Import

i-Import is an online application which allows importers to apply for and receive permits to import medicines online and EFDA staff to manage these applications online.

[VIEW DETAILS](#)

Welcome to EFDA! Our Vision i-Import i-Register i-License Traceability

[Privacy - Terms](#)

MoH: <https://www.moh.gov.et/ejcc/en>

አማርኛ | Afaan Oromoo | ትግርኛ | English

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<https://www.moh.gov.et/ejcc/am/covid19-courses> More than 16 ከ16 ሚሊዮን ብር በጤናው ዘርፍ የላቀ

EPSA: <https://epsa.gov.et/>



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With over seventy years of service, the Ethiopian Pharmaceuticals Supply Agency (EPSA) is a veteran institution in providing pharmaceuticals, laboratory reagents, medical equipment and supplies to the Ethiopian people. Despite the numerous changes in nomenclature throughout the years, supplying pharmaceuticals has always been our core activity. Considering the population size has grown in folds since our establishment, we have launched an all rounded reform in our work procedures. Equipped with revamped work procedures, visionary leadership and dedicated staff, we believe the future is bright for pharmaceuticals supply in Ethiopia.

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Suppliers perception survey questionnaires:

[Supplier Perception Survey \(surveymonkey.co.uk\)](#)

EPSA Ethics survey questionnaires:

[Assessment of Corruption and Malpractice in](#)

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About AHRI



Background

The Armauer Hansen Research Institute (AHRI) was founded in 1970 through the initiative of the Norwegian and Swedish Save the Children organizations seconded by the Ministry of Health of Ethiopia.

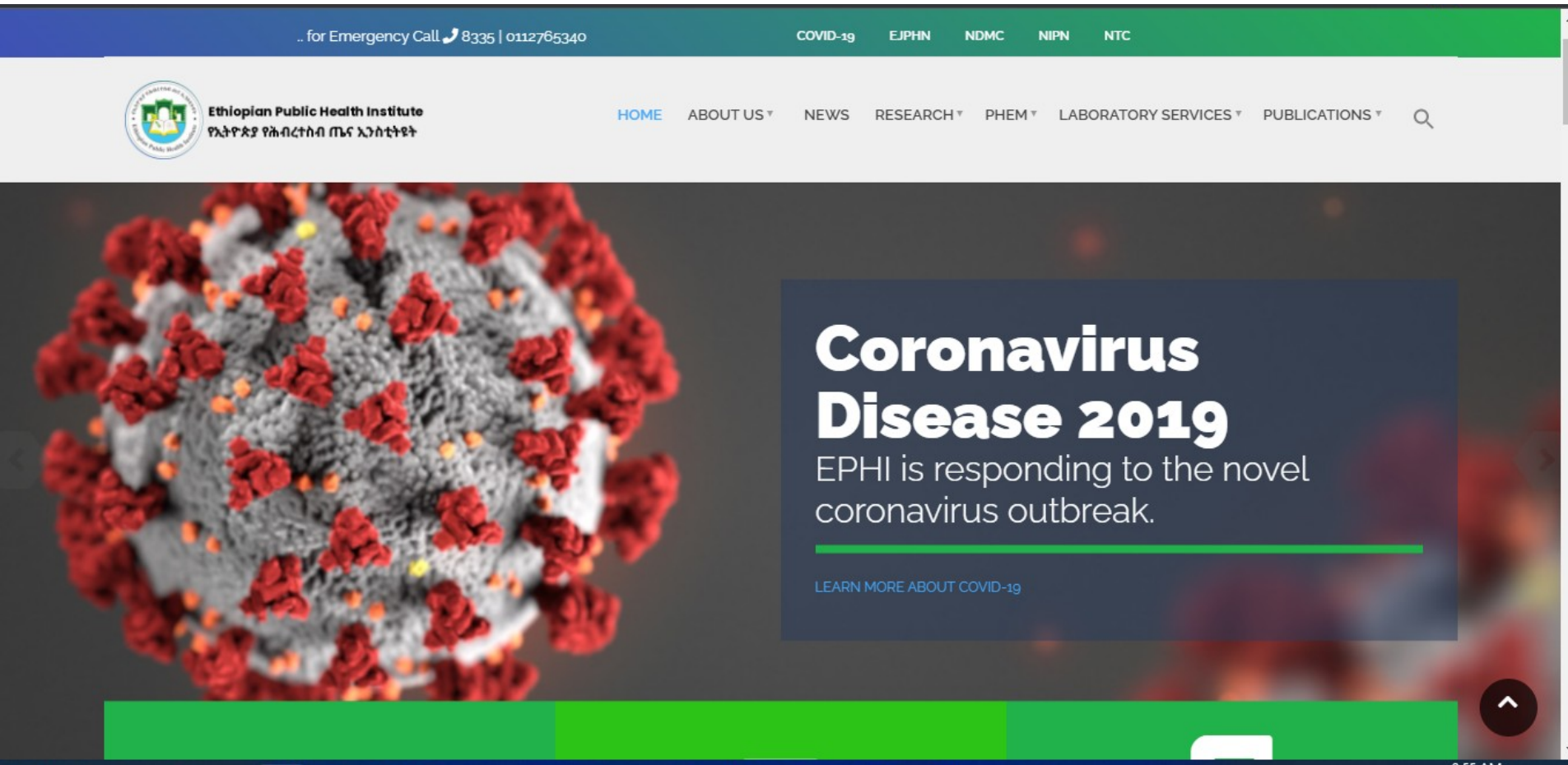
The Institute got its name from the Norwegian physician, Gerhard Henrik Armauer Hansen, who first described the leprosy bacillus (*Mycobacterium leprae*).

AHRI was established as a biomedical research institute located next to the All Africa Leprosy Rehabilitation and Training Hospital (ALERT). The Institute joined the Ethiopian Ministry of Health in 2004.

AHRI receives core support for its research activities from [SIDA](#) and [NORAD](#). It enjoys tax free privileges from the Ethiopian government. Postgraduate (MSc and some PhD) theses are funded from the core budget of the Institute. Most of other research is funded by competitive grants. AHRI research activities cover basic (immunology and molecular biology), epidemiological and translational research. AHRI has published more than 900 papers in peer

9:51 AM

EPHI: <https://ephi.gov.et/>



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EMERGENCY

Coronavirus disease (COVID-19) pandemic

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Coronavirus disease (COVID-19) Situation dashboard

This interactive dashboard/map provides the latest global numbers and numbers by country of COVID-19 cases on a daily basis.



Timeline: WHO's COVID-19 response

Explore the interactive timeline showcasing how the organization has taken action on information, science, leadership, advice, response and resourcing.

CDC: <https://www.cdc.gov/>

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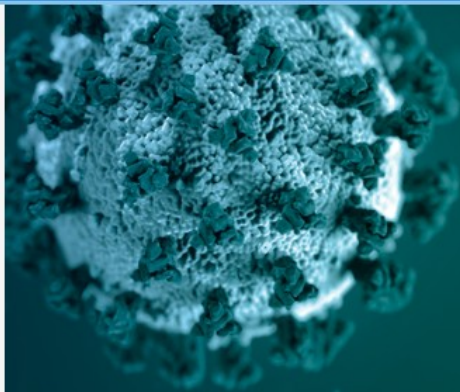
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Coronavirus Disease 2019

CDC is responding to the coronavirus outbreak.

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8:52 AM

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COVID-19



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November is National COPD Awareness Month

The work you do can affect your lungs, but you can protect yourself from chronic obstructive pulmonary disease.

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Coronavirus (COVID-19)

[Get the latest advice about COVID-19](#)

A background image showing healthcare workers in a clinical setting. On the left, a person in a white lab coat is partially visible. On the right, a person in blue scrubs with a stethoscope around their neck is writing on a clipboard. The image is partially obscured by a blue text box on the left.

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Quality of Web sites

- ❖ **High Quality:** Government (.gov), education (.edu), professional association (.org), military (.mil); news services
- ❖ **Medium Quality:** Pharmaceutical manufacturers (.com)
- ❖ **Low Quality:** individuals, selling ideas or products(.com)

Web Sites...

- three - letter extension
 - [.edu](#): An educational institution
 - [.biz](#): business
 - [.com](#): commercial (e.g., <http://www.microsoft.com> or <http://www.netscape.com>)
 - [.gov](#): government (e.g., <http://www.cdc.gov> for the CDC)
 - [.info](#): information
 - [.mil](#): military
 - [.org](#): organization
 - [.net](#): network (network provider)
- Two - letter code that indicates the country
 - [.fr](#) = France;
 - [.us](#) = United States
 - [.et](#) = Ethiopia

Literature Evaluation : Controlled Clinical Trial Evaluation

Chapter 4

Learning objectives

- **At the end of this session, the student will be able to:**
 - 1) Describe special characteristics of a **controlled clinical trial**
 - 2) Describe each criteria used to appraise each section of a clinical trial
 - 3) Differentiate between internal and external validity
 - 4) Discuss the need for a well-defined study objective, randomization, and blinding within a clinical trial
 - 5) Determine statistical significance, clinical difference, and clinical meaningfulness of the clinical⁷⁷ trial results

Introduction

- The practices of medicine and pharmacy are dynamic, and **drug facts acquired during formal education cannot sustain** a health care provider in future practice.
 - ✍ Changes include *new medications, dosage formulations* and *uses approved by the FDA*, *revised drug safety information* (i.e., adverse drug effects, drug interactions), and *updated disease state therapeutic guidelines*.
- In 2024, 47 new molecular entities/biological agents and 50 novel drugs were approved by the FDA.

Introduction

- Pharmacists continually rely upon the biomedical or pharmacy literature for many day-to-day activities
- Pharmacists need to efficiently locate, critically analyze, and effectively communicate data from the primary literature in daily activities of patient care and the medication use process
 - ✓ Therefore, skills such as drug literature evaluation are necessary to prepare the HCPs for practice

Introduction

- Multiple resources are available for pharmacists to provide answers to questions, care for patients, make decisions, and solve problems
- Pharmacists need:
 - ✓ To recognize both the **advantages** and **limitations** of the information resources
 - ✓ To **clarify, explain, defend**, and/or **refute** information
 - ✓ To have skills in efficiently locating and critically analyzing drug information

Biomedical/Pharmacy Literature

Three Types of Literature

Literature Type	Description	Examples
Tertiary	Established knowledge	Textbooks, review articles, MD Consult, Lexicomp
Secondary	Indexing/abstracting services (databases)	- PubMed or MEDLINE (National Library of Medicine), - Embase (Elsevier) - IPA (Thomson Scientific Inc.) CINHAL
Primary	Original research	Controlled clinical trials, case-control studies, crossover trials, case reports

RCTs

- Primary literature, specifically controlled clinical trials, serves as the foundation for clinical practice
 - ✓ **Are a reliable source of new information** to change health care practices
 - ✓ A controlled clinical trial is the *premier study design to measure and quantify differences in effect of the intervention versus control*

RCTs

- In general, a **controlled clinical trial** consists of an *investigational (intervention) group* being directly compared to a *control group* (e.g., standard therapy, placebo)
- The **intervention** under investigation may be
 - ✍ new medication,
 - ✍ different medication dosing regimen,
 - ✍ diet,
 - ✍ surgery,
 - ✍ behavioral process,
 - ✍ exercise program,
 - ✍ diagnostic procedure, or something else.

RCTs

- The goal of the clinical trial is to measure and quantify the difference in **effect** (efficacy, safety) between the investigational and control groups.
- The results then can allow decisions to be made regarding proper care for patients (i.e., *to use or not use* the investigational therapy).
- The clinical trial should be **presented** in a manner that explains the **research process** in an orderly format to improve the readers' comprehension of the **project, results, and conclusions**.
- The **CONSORT** (**C**onsolidated **S**tandards **o**f **R**eporting **T**rials) format was formulated to improve the quality of reporting clinical trials in the published literature.

Format and Content of Controlled Clinical Trials

Section	Type of Information Presented
Abstract	Brief overview of the research project
Introduction	Research background What is already known Needs for the study/rationale Objective/hypotheses
Methods	Study design Study Setting Population to be sampled Inclusion and exclusion criteria Intervention and control groups Randomization Blinding Endpoints/outcomes Follow-up procedure Sample size calculations/Power analysis Statistical analysis

Format and Content of Controlled Clinical Trials

Section	Type of Information Presented
Results	Subject characteristics Subject dropouts/adherence Endpoints quantified Statistical significance Safety assessments
Discussion	Result interpretations Clinical significance Other study results compared Limitations Conclusion/Application to practice
Acknowledgments	Other contributors Funding source Peer-review dates/Manuscript acceptance date (not all trials)
References/Bibliography	Citations for information included from other resources (e.g., trials and reports)

Critical literature evaluation

- **Critical literature evaluation** is the process of reading and evaluating primary literature journal articles in order to arrive at an interpretation that you can call "*your own*"
 - ✓ Are the results **believable?**
 - ✓ Are the results **applicable to your practice?**
 - ✓ Does the **paper really support its claims?**
- The ultimate interpretation and decision about the value of an article rests with **the reader**
- Many studies are open to differing interpretation
- The majority of clinical decisions are based on primary literature reports

Critical literature evaluation

- Literature evaluation skills are needed when
 - ✓ Therapeutic dilemmas arise
 - ✓ When information about a particular subject is conflicting
 - ✓ When making decisions about drug policy (i.e., adding or deleting a drug from an institution's drug formulary)
- It has been estimated that **40 to 50%** of published articles in the medical literature have **serious problems** with study design, statistical analysis, and conclusions

Critical literature evaluation

- Reasons for publications of poor quality research
 -  The "Publish or Perish" dilemma
 -  Investigator's lack of knowledge in study design or statistical analysis
 -  Peer-reviewer's lack of knowledge in study design or statistical analysis
 -  Preliminary results published to stimulate further research in area
 -  Scientific fraud

Approach to Evaluating Research Studies

Journal, **Peer-review** and Investigators

- ❖ One essential journal feature is the **peer-review process**
- ✍ Peer-reviewed articles are evaluated by someone other than the editorial staff
- ✍ Reduce the publication of manuscripts that have inappropriate methods/design, are poorly written, and/or do not meet the needs of the journal's audience
- ✍ Does not always prevent publication of articles with deficiencies
- ✍ Check journals for information addressing whether the peer-review process is used or not

- **Certain journals have a reputation for good-quality publications**
 - ✓ *New England Journal of Medicine (NEJM) and Annals of Pharmacotherapy*
 - ✓ This too can be considered in the evaluation of literature, although poor articles can still be found in **well-respected** journals, and excellent articles are published in other journals

Evaluating the Journal

- **Characteristics of Reputable Scientific Journals**
 - ✓ **Editorial policy** that specifies the requirements for the types and formats of submitted manuscripts
 - ✓ **Peer-review policy** which requires that all manuscripts be reviewed by consultants or researchers in the field of study prior to publication
 - ✓ **An editorial board** that is composed of well-known researchers and leaders in their respective discipline (editorial board is usually listed on or near the title page)
 - ✓ Does not contain an **overabundance of advertisements**
 - ✓ Reputable journals usually publish **supplements**.
 - ✓ Keep in mind that information found in supplements may not be peer-reviewed

Approach to Evaluating Research Studies

- **Investigators' credentials** and the practice site of these individuals
 - ✓ need to be properly *trained*
 - ✓ have active practice *experience* in the area of study
 - ✓ should be a site that has the *capability to perform the study* (i.e., have the resources to properly and completely perform the necessary study methods)
 - ✓ Persons with specialized credentials in *biostatistics* need to contribute with statistical analysis of the data
 - ✓ all authors listed should have made *substantial contributions* to the research and/or publication

Evaluating the Investigators

- The investigators should have a ***good track record*** of prior research
- ***A biostatistician*** should be involved with the evaluation of data
- The validity of studies authored ***entirely by investigators*** working in ***pharmaceutical industry*** is sometimes questioned because the goal of these individuals is primarily promotion of the company's drug products

Evaluating the Investigators

- The **source of funding** and *potential conflicts of interest* should be disclosed by the authors.
- Complete funding by pharmaceutical companies may be an issue of concern.
- However, potential bias does not necessarily negate the study results
- The research site should have appropriate **resources and technology** to effectively conduct the study

The Clinical Trial Title

- ✍ A title should be reflective of the work, unbiased, specific, and concise (i.e., usually ≤ 10 words), but not too general or detailed
- ✍ **Declarative sentences**, which tend to overemphasize a conclusion, are ***not preferred for scientific articles***
- ✍ The title should not be phrased as a question
- ✍ Randomized clinical trials should be identified in the title
- ✍ The title should include terms both **sensitive** and **specific** that allow electronic retrieval of the article

The Clinical Trial Title

- "Improved bronchodilation with levalbuterol compared with racemic albuterol in patients with asthma."

Vs

- "Bronchodilation from levalbuterol compared with racemic albuterol in patients with asthma: a randomized clinical trial."

Abstract

- Considered to be a **concise overview** of the study or a synopsis of the major principles of the article
- *Decision-making should not rely solely on reading abstracts; the entire manuscript is to be read and thoroughly evaluated*
- Abstracts include information addressing the article **objective, methods, results, and conclusions**
- A primary use of abstracts is for readers to obtain an immediate overview of the article to determine if the entire article should be read
- Another use is publishing the abstract in secondary resources (e.g., PubMed, IPA for individuals conducting literature searches)

Abstract....

- The structured abstract includes the following sections: usually contain **< 500 words**
 - ✓ Objective, research design, clinical setting, participants, interventions, main outcome measurements, results, and conclusions
- Abstracts should be consistent with the manuscript and should not present biased and/or inaccurate information

Abstract....

- Structured abstracts, compared to unstructured abstracts, do have some advantages, including being more informative, easier to read, and generally preferred by readers.
 - However, structured abstracts usually require more journal space.
- Informative abstracts may entice some individuals to read the study; thus, abstracts should be thorough, complete, and unbiased in wording selection

Introduction

- The introduction section serves two specific purposes:
 - ✍ Discussing the **study rationale** and **study purpose**
- The study investigators may state that the reason the research was conducted was **due to the lack of data to answer a question** or **because available data are conflicting regarding an issue**.
- The study objective is often stated within the **last paragraph**
- **Well-written studies present a clearly stated research purpose, and this statement should be understood by the reader before continuing with the remaining article content**
- Studies with a well-written purpose statement enable the reader to better comprehend and assess the research methods

Introduction....

- The investigators need to **formulate a research and null hypothesis**
 - ✓ A **research hypothesis** (also known as the alternative hypothesis) is a ***difference between the therapy under investigation and the control*** while the **null hypothesis** (H₀) is ***no difference between these two groups***
- Readers should recognize that **not all clinical trials will include the specific research and null hypotheses** in the introduction section
 - ✓ this can be considered a deficiency in the paper, but does not necessarily indicate that the paper contains incorrect information

Introduction...

- The Improved Reduction of Outcomes:Vytorin Efficacy International Trial (IMPROVE-IT), Ezetimibe added to statin therapy after acute coronary syndromes
 - compared simvastatin 40 mg plus ezetimibe 10 mg or simvastatin 40 mg plus placebo
 - the rationale for conducting this clinical trial was that no evidence has been published that determines whether **ezetimibe** added to **simvastatin** can improve CV events
 - The research hypothesis of IMPROVE-IT:“**There is a difference in the incidence of CV events in patients receiving combination therapy with ezetimibe and simvastatin compared to simvastatin monotherapy.**”
 - The H0 for this clinical trial was:“**There is no difference in the incidence of CV outcomes between combination therapy with ezetimibe and simvastatin compared to simvastatin monotherapy**”.

Introduction

- Authors may set the stage by **presenting only selective** (i.e., not comprehensive) information and/or weak references to support the rationale for conducting the study
- The information may be presented using **biased wording**, which predisposes the reader to believing the prior research was insignificant in providing evidence applicable to practice

Methods

- The methods section of a clinical trial contains a large amount of information that includes:
 - ✓ the type of ***subjects enrolled***, the ***comparative therapy description, outcome measures***, and ***statistics***
- The ***design of a study*** (i.e., methods) is important for the results to be valid
- Flaws within the design of a clinical trial limit the application and significance of the results
- Poor study design leads to reduced ***study internal validity***, thus resulting in ***limited external study validity***

Methods: Validity

- **Internal Validity:**

- ✓ Within the confines of the study, the **results appear to be accurate**, the **methods** and **analysis** are appropriate and the **interpretations** of the investigators appear supported

- **External Validity:**

- ✓ The conclusions of the study can be applied to the ***reader's practice***. Also referred to as “***generalizability***” of results

Methods: Validity...

Internal vs. External Validity of Clinical Trials

Term	Meaning	Application
Internal validity	Quality of the study design	Strong design should translate into reliable results
External validity	Ability to apply results into practice	Study results meaningful to practitioners and can be used for patient care

Devote the majority of your time used to assess the trial in the method section

Methods: Study Design...

- Identify the strengths and limitations of the research designs
- Controlled clinical trials offer investigators the most rigorous method of establishing a *cause-and-effect* relation between treatment and outcome

Methods: Patient Inclusion/Exclusion Criteria

 *The results of a controlled clinical trial should be extrapolated to the patient type enrolled in the study*

The inclusion criteria

 lists subject demographics that must be present to be enrolled into the trial

 the inclusion criteria attempt to include subjects that are homogeneous and are similar to the common type of patients in practice

Exclusion criteria

✓ are characteristics that prevent enrollment into the trial or necessitate withdrawal from the study, if they are later determined to be present

Methods: Patient Inclusion/Exclusion Criteria

- ✍ The inclusion/exclusion criteria are pertinent to the extrapolation of the study results
- ✍ Researchers are careful in deciding which subjects to include and exclude in the clinical trial
 - ✓ Standard types of subjects disqualified are **pregnant and lactating** females; also, most clinical trials will not enroll subjects with severe conditions

Methods: Patient Inclusion/Exclusion Criteria

- The investigators of IMPROVE-IT enrolled men and women with acute coronary syndrome (an acute myocardial infarction [MI], with or without ST-segment elevation on electrocardiography, or high-risk unstable angina).
- Additionally, patients were at least 50 years of age and were required to have an LDL-cholesterol level of 50 mg/dL.
- Thus, the results of IMPROVE-IT cannot be extrapolated to all patients who are younger than 50 years of age since only those who were at least 50 years of age with acute coronary syndrome were included in this clinical trial.

Methods: Patient Inclusion/Exclusion Criteria

- During the process of the investigators selecting subjects to be included into a study, readers of clinical trials need to be conscious of the potential for a **selection bias** that may be present.
- A selection bias occurs after subjects meet the inclusion and exclusion criteria, but are not enrolled into the study
- One common form of a selection bias is requiring the subjects to complete a **run-in phase** (also called **lead-in phase**) before being officially enrolled in the study
- This phase is usually short in duration (usually 2 to 4 weeks) in which the subjects may take a placebo or the therapy being investigated
- Typical reasons include **identifying subjects that may or may not adhere with the therapy regimen, experience side effects from the therapy, or did not meet prespecified criteria**

Methods: Patient Inclusion/Exclusion Criteria

- Afterwards, these identified subjects are excluded from participating in the study even though they met the original inclusion criteria.
- The run-in phase produces a bias by selecting a group of subjects who do not completely represent the population, **since a selected group of the subjects meeting the study inclusion criteria are not included** in the study and their run-in phase results are not included in the final analysis
- Researchers may include a run-in phase in which only those persons achieving a preset goal are allowed to be included in the study.

Methods: Patient Inclusion/Exclusion Criteria

- Investigators also should explain the process of recruiting subjects and define the time period of which the recruitment occurred.
- The methods in which investigators recruit subjects may have implications on the generalizability of the research results to the population (i.e., external validity).
- Newspaper and Internet advertisements are common; however, there are inherent problems with this form of advertisement.
- The majority of persons who read the newspapers are older in age, Caucasian, wealthier, more educated, and own more upscale homes than the average American.
- Gender bias also can affect recruitment rates as it has been documented that female readers consider newspaper advertising to be more important than do male readers

Methods: Patient Inclusion/Exclusion Criteria

- **Minority** and **elderly** individuals are less familiar and have less access to the Internet.
- Subjects registering via the **Internet** were significantly younger than those registering through the call center

Methods: Intervention and Control Groups

- ✍ The **intervention group** consists of the new therapy under investigation (e.g., medication, procedure)
- ✍ The **control group** can consist of no therapy (e.g., **placebo**), **another therapy** (a.k.a. active control) for instance, drug or exercise, or **compared to existing data** (i.e., historical data)
- ✍ Readers should be aware that clinical trials with a **placebo as the control**, particularly those funded by the pharmaceutical industry, yield a **larger treatment effect** than if an active therapy was selected as the control group

Methods: Intervention and Control Groups

- ✍ One advantage of using historical controls is that **only one group** is needed to be enrolled, which may result in **less time, expense, etc.**
- ✍ Another advantage is the usefulness of studying a **disease with a low occurrence** or a disease with high incidence of death or other serious sequelae in which **some form of therapy should not be denied.**
- ✍ Disadvantages include usually **overestimating** the effect of the intervention, **lack of homogeneity** between patients in the trial and historical controls, and **differences in therapeutic procedures or techniques** from one study period to the next

Methods: Intervention and Control Groups

- ✍ Historical controls are not used very often in published clinical trials, but are acceptable in selected situations.
- ✍ For example, the use of a placebo group is not considered ethical in clinical trials evaluating new therapies in patients with **epilepsy**.
- ✍ Due to the morbidity and/or mortality associated with an uncontrolled seizure disorder, this patient type should not be denied therapy in order to investigate a new therapy.

Institutional Review Board (IRB)/Subject Consent

- ✍ Research projects that use **humans as study subjects** must be approved before investigators begin enrolling subjects into the trial
- ✍ The IRB is the committee charged with ensuring the **subjects are protected and not exposed to unnecessary harm or unethical medical procedures**
- ✍ Another primary responsibility of the IRB is to approve the **informed consent form**
- ✍ RCT should include a statement that the research was approved by the IRB (or other committee that protects subjects) and consent was obtained from the subject to participate in the research project

Methods: Blinding

- ✍ Blinding techniques are incorporated to *reduce possible bias* (defined as “differences between the true value and that actually obtained [are] due to all causes other than sampling variability”)
- ✍ Blinding is a technique in which subjects and/or the investigators are unaware of *who is in the intervention or control group*.
- ✍ This is especially important in studies measuring *subjective outcomes* (e.g., pain, depression scores)
- ✍ Patients knowing they are taking a placebo to reduce depression symptoms are likely to report no change or worsening of the disease

Methods: Blinding

- ✍ Blinding techniques are important to reduce the influence of bias on measuring a difference in effect between the intervention and control
- ✍ **Double-blinding** is considered the gold standard blinding technique and is most commonly used in clinical trials

Methods: Blinding

Type of blinding	Definition
No blinding (open-label)	Investigators and subjects are aware of the assignment of subjects to the intervention or control group
Single-blinding	Either investigators or subjects , but not both, are aware of the assignment of subjects to the intervention or control group
Double-blinding	Both investigators and subjects are not aware of the assignment of subjects to the intervention or control group
Triple-blinding	In addition to both investigators and subjects not being aware of the assignment of subjects to the intervention or control group, trial personnel involved with data analysis and interpretation are not aware of subject assignment

Methods: Blinding

- Single- and non-blinding techniques are primarily incorporated in clinical trials that have **study objectives not conducive to blinding** (e.g., surgery versus medication).
- Some trials may include a procedure that is difficult to blind (e.g., surgery)
 - It may not be ideal to include a placebo procedure because **sham surgery** is not without risks as **death or infection-related complications** are possible

Methods: Blinding

- An example of **single-blinding** of investigators is when one group of subjects is administered a medication **subcutaneously once daily** versus the other group who took an **oral anticoagulation medication**.
- The subjects would not be subjected to blinding:
 - Since the risk of injecting a saline solution subcutaneously (i.e., bleeding complications may develop in persons taking an anticoagulant agent plus unnecessary injections) may outweigh the benefit.
- The investigators would measure the occurrence of a **blood clot**, an objective outcome that cannot be biased or influenced by the subjects.
- Therefore, the subjects' knowledge of which therapy they were receiving cannot influence the incidence of the blood clots.
- Since the investigators do not know which therapy is administered, the potential for the results to be biased is minimized.
- There are also cases in which the subjects were blinded, but not the investigator.

Methods: Blinding

- **Double-blinding** is considered the gold-standard blinding technique and is most commonly used in clinical trials.
- As a general rule, and regardless of whether the outcome is a subjective or an objective measure, the study should be double-blinded.
- A clinical trial measuring an objective outcome usually assesses other study outcome measures, such as the **incidence and severity of side effects**, which may be biased if double-blinding was not incorporated into the trial.
- For instance, double-blinding was used in IMPROVE-IT to minimize biases in not only **objective measurements** (liver enzyme levels and creatine kinase levels) and also **subjective assessments** (side effects) in patients assigned to intervention or control

Methods: Blinding

- In order to *ensure that blinding remains intact*, the therapy each group receives should be exact in *frequency of administration, appearance, size, taste, smell*, and *other variables*.
- Studies that compare regimens taken *once daily* to *twice daily* will require the once-daily group to take a placebo as the second dose (i.e., *double-dummy*).
- Double-dummy methods are included in clinical trials when two therapies being compared are *not the same* (different routes of administration, different formulations).
- Patients receive *two formulations*, one active and one control, to ensure that blinding is maintained

Methods: Blinding

- For example, investigators of a clinical trial evaluating the blood pressure lowering effects of *amlodipine* (a tablet) and the combination product of *amlodipine plus benazepril* (a capsule)
 - Should administer amlodipine tablets plus placebo capsules to those subjects randomized to amlodipine therapy and amlodipine/benazepril capsules plus placebo tablets to the other subjects
- Sometimes it is necessary to triple-blind a study.
- In addition to the trial investigators and subjects, other personnel involved with the trial
 - Data collection, analysis, or monitoring; drug administration or dispensing
 - Can have opinions regarding the outcome of the therapy being studied based on their interaction with the subjects involved in the trial or their experience with the intervention and/or control being assessed.
 - These opinions may cause inappropriate data collection, measurement, analysis, and/or interpretation of the results by the study personnel.

Methods: Randomization

- ✍ Ensures that "all study subjects have an equal chance of being 'assigned' to either the intervention or control group."
- ✍ **Randomization** is a distinguishing study attribute that separates controlled clinical trials from other study designs (e.g., case-control, cohort).
- ✍ Research has indicated that the results obtained from randomized trials are **more dependable** than nonrandomized trials.
- ✍ Subjects are randomized so the investigators cannot purposely assign selected persons to one group over another
 - ✍ i.e., individuals with **more comorbidities in the control** versus **healthier individuals in the intervention group**.

Methods: Randomization

- ✍ Randomization minimizes bias by lowering the potential for an **imbalance of risk factors or prognostic variations** between the intervention and control groups
- ✍ A difference in effect measured by a clinical trial may result from many causes, and treatment may be just one of these.
- ✍ Disparities between the groups at baseline may cause a false result instead of measuring differences in effect between the intervention and control.
- ✍ Therefore to be assured that the difference is truly due to the intervention, the groups need to be as similar as possible to control for any confounding variables

Methods: Randomization

- ✍ Many randomization techniques are available and range from very simple to complex processes.
- ✍ Specific randomization methods include simple (i.e., coin toss), blocked, and stratified (more advanced).
- ✍ **Simple randomization** is an easy technique to implement and includes assigning subjects according to some criteria (e.g., day of the week, subject birthday, or subject medical record number)
- ✍ But this method is not considered a proper randomization method since the **number of subjects in the groups can be imbalanced** due to the technique.
- ✍ If investigators assign all subjects with an office visit on a specific day of the week to one group (i.e., control), then these subjects did not have equal opportunity to be assigned to either group.

Methods: Randomization

- ✍ **Stratification** is designed to achieve similarities in both known and unknown baseline patient characteristics between the groups.
- ✍ Selected factors are identified (age, smoking, presence of other disease states)
 - ✍ and used in determining which group the subjects will be assigned so significant imbalances of these factors are not present among the groups, while all subjects with any specific factor have an equal chance of being in each group.

Methods: Randomization

- ✍ For example, the risk of developing a venous thrombosis is greater in cancer patients who are **immobilized and/or receiving hormonal therapy**.
- ✍ Thus, a study comparing two agents (e.g., LMWH vs UH) to prevent a venous thrombosis in this patient type would use stratified randomization to ensure a similar number of patients with these two risk factors are assigned in each of the two study groups.
- ✍ If one group had significantly more patients taking hormonal therapy, this factor could be the reason more patients had a venous thrombosis instead of the drug not being effective as the other drug.
- ✍ Thus, a false study result could occur due to the two patient groups not being similar at the beginning of the study.

Methods: Endpoints

- ✍ All trials specify one effect caused by the intervention and control as the primary endpoint:
 - ✓ Which can be referred to as "what did the investigators measure to achieve the study objective"
 - ✓ The controlled clinical trial primary endpoint should be appropriate for the study purpose and measured using valid techniques and methods
- ✍ Researchers also usually measure a primary endpoint plus secondary endpoints.
- ✍ These secondary endpoints are important, but not considered to be the primary purpose of the study

Methods: Endpoints

- ✍ For example, a trial evaluating the cholesterol lowering effect of an HMG coenzyme A reductase inhibitor compared to placebo selected a **change in average LDL-C value**, an appropriate measure, as the **primary endpoint** to satisfy the study objective.
- ✍ However, measuring a **change in serum creatinine** between losartan and lisinopril to improve heart failure (HF) symptoms is not ideal, since serum creatinine is not the predominate parameter used in practice to monitor the progression or improvement in HF status.

Methods: Endpoints ...

- ✍ Investigators may combine a group of endpoint measures into one primary endpoint, referred to as a composite endpoint.
- ✍ The group usually consists of clinical outcomes directly related to morbidity and mortality as opposed to a pharmacological action (e.g., reduction in any incidence of stroke/MI/cardiovascular-related death versus lowering cholesterol levels).

Follow-Up Schedule/Data Collection/Compliance

- ✍ A study should be conducted for an appropriate duration and data need to be consistently collected throughout the entire trial.
- ✍ A magical number of weeks or months has not been established as a rule for all clinical trials, but the length of the study (i.e., follow-up time) should be an ideal representation to answer the question being researched.
- ✍ Statins usually exert the maximum cholesterol lowering effect after approximately 6 weeks of stable dosing.
 - ✍ Thus, the results of a study directly comparing the effects of atorvastatin 10 mg once daily to simvastatin 20 mg once daily for 6 weeks on the LDL-C level lowering differences between these two agents would be considered acceptable

Follow-Up Schedule/Data Collection/Compliance

- ✍ Data collection methods should be reasonable, in that extensive time and/or procedures are not required
- ✍ Measuring the compliance of therapy in the study participants
- ✍ Subjects not complying with the therapy regimen may cause inaccuracies and less reliable data
- ✍ Insufficient and/or inappropriate data collection methods and noncompliance usually lead to biased results that may make the extrapolation of results to clinical practice difficult
- ✍ A number of trials do not have an extensive follow-up time and the reader may have difficulty in interpreting the results for clinical practice

Methods: Sample Size

- ✍ Sample size is of considerable importance to the validity of the study results
- ✍ Financial and logistical limitations prevent all subjects with the specific inclusion criteria from being enrolled into the study
- ✍ The number of subjects to enroll in a clinical trial is dependent on the expected magnitude of difference in the endpoint effect between the intervention and control

Methods: Sample Size

- ✍ An inverse relationship exists between the sample size and the effect size
- ✍ Sample size must be calculated prior to initiating the clinical trial
- ✍ Sample size is calculated based on the differences in the primary endpoint effect between the intervention and control groups

Methods: Statistical Analysis

- ✍ Purpose is to collect sufficient evidence to reject the null hypothesis (H_0) in favor of accepting the research hypothesis (H_1)
- ✍ Prior to starting the study, appropriate tests are selected based on the type of data that will be collected and analyzed

Results

- ✍ This section contains primary and secondary endpoint results and other useful information.
- ✍ Should be critically appraised to verify if the study objective was met, based on the data, and to evaluate the other types of outcomes that may have occurred.
- ✍ Normally tables, charts, figures, or other illustrative forms present many of the results, which can expedite the understanding and analysis process.

Results: Subject Demographics

- ✍ Describes the subjects actually enrolled and randomized in the clinical trial.
- ✍ A general overview of the average subject is described, usually presented in a table of baseline information.
- ✍ Typical information in the table includes average age, male to female ratio, disease states, and/or drug therapy among the study participants at the time of study enrollment.

Results: Subject Demographics

- ✍ Patient baseline demographic data need to be compared between treatment groups to ensure that the groups are as similar as possible.
- ✍ Groups should not have any significant differences if proper randomization techniques are incorporated by the study investigators; but a few differences can still occur due to chance.

Results: Subject dropout and compliance

- ✍ Data regarding the follow-up (i.e., subject dropout) and compliance should be presented.
- ✍ Not all subjects randomized in a clinical trial will complete the entire duration
- ✍ Also, not all subjects will be compliant with the therapy.
- ✍ Investigators need to report the number of subjects and major reasons for discontinuing the study.

Results: Subject dropout and compliance

- ✍ The impact of attrition on the overall study results is dependent on the magnitude of subject discontinuations.
- ✍ A few subjects dropping out of the study may not cause a substantial difference in the results, whereas a sizable percentage of attrition may alter the study results significantly.
- ✍ **Intention-to-treat (ITT) analysis:** study results will be analyzed using data collected from all randomized subjects, regardless of whether they completed the entire study duration.
 - ✍ Results from dropouts are not discarded, but are considered to be **treatment failures**
 - ✍ Even in cases where the subject may not have taken even one dose of the medication under investigation, these results are still included in the ITT analysis
 - ✍ “Once randomized, always analyzed”
 - ✍ ITT analysis better **mimics real-life application** of an intervention into practice.

Results: Subject dropout and compliance

- ✍ **Per-protocol (PP) procedure:** analyzing data only from subjects completing the trial.
- ✍ PP procedure helps determine the effects of the intervention in subjects that followed the study protocol and completed the entire course of therapy.
- ✍ The ITT analysis can be thought of as a **worst-case scenario model**, whereas the PP analysis can be thought of as a **best-case scenario model**
- ✍ Optimally, study investigators will present both ITT and PP results; the desire is to see no significant differences between the ITT and PP results.

Result: Endpoints/Safety

- ✍ The information should be presented clearly and completely, using clear and unbiased methods.
- ✍ Investigators need to explain the results and present **p-values**.
- ✍ Results of secondary endpoints follow and are presented in a fashion similar to the primary endpoints.
- ✍ Endpoints other than the primary endpoint may not be adequately powered to detect differences in effect between the intervention and control.

Results: Endpoints/Safety

- ✍ Safety assessments or tolerability of all therapies should be included in the results section.
- ✍ Investigators need to implement **valid methods** of defining, collecting, and analyzing these results.

Results: Surrogate Endpoints

- ✍ A measure of the efficacy of a treatment can be defined as:
 - ✓ laboratory values (e.g., HDL-C/LDL-C),
 - ✓ symptoms (e.g., pain), or
 - ✓ clinical parameters (e.g., blood pressure), which are employed as **a substitute for a clinical endpoint** (e.g., morbidity, mortality)
- ✍ Assumption is that changes in **the surrogate endpoint** can be directly translated into changes in the **definitive clinical endpoint.**
- ✍ Surrogate endpoints are selected for clinical trials is to **quickly measure an effect** at a lower expenditure.

Results: Surrogate Endpoints

- ✍ Ideal surrogate endpoint (to be considered as a valid substitute for a clinical endpoint):
 - ✓ convenience (easily and readily assessable);
 - ✓ well-established relationship between the surrogate and clinical outcomes; and
 - ✓ determination of clinical benefit as a result of changes in the surrogate endpoint.

Results: Subgroup Analysis

- ✍ Analysis of the results of **subsets** of the study subjects as divided into various groups that often include **gender**, **age**, and **presence of diseases** or other complicating factors (i.e., diabetes vs. no diabetes).
- ✍ Reasons to analyze the results in these subgroups vary, but usually relate to **providing additional information** in these specific patient types **as opposed to just the overall trial results** from all the randomized subjects.

Results: Subgroup Analysis, Limitations

- ✍ Subgroup analysis results from flawed studies may be of no importance.
- ✍ Investigators may conduct a multitude of subgroup analyses and only report the statistically significant results.
- ✍ Power of the assessment is reduced, since results from a smaller number of subjects are analyzed as compared to the entire trial sample.
- ✍ The primary endpoint should be statistically significant before subgroup analyses are conducted; otherwise investigators may be searching for statistically significant results.

Results: Subgroup Analysis, Limitations

- ✍ The subgroup analyses should be defined prior to the initiation of the trial and have documented justification to be conducted (e.g., past studies results suggest an effect in this group).
- ✍ Outcomes that can be influenced by either the intervention or control (i.e., compliance) should not be selected for a subgroup analysis.
- ✍ The reader needs to review appropriate subgroup analysis, but keep the primary endpoint as their focus.

Results: Ancillary versus Adjunctive Therapies

- ✍ Clinical trials are designed to have almost identical groups with the only difference between the groups being the assignment to the intervention or control.
- ✍ At times, the study design may allow **ancillary therapy** to be included, in which subjects can take another therapy that can distort or confound interpretation of the results.
- ✍ The effect of the ancillary therapy on the study results needs to be assessed and included in the study evaluation.
- ✍ Some studies may include in the methods an **adjunctive therapy**, which **all participants receive as treatment**, while ancillary therapy may not be equally distributed between the intervention and control groups.

Discussion/Conclusion

- ✍ The primary purpose of discussion/conclusion is to **evaluate and interpret** the results of the clinical trial.
- ✍ Investigators should begin with a summary of the key findings of the study. Potential explanations of the study results should be addressed.
- ✍ Should discuss the internal validity and external validity.
- ✍ Should interpret the trial results in comparison with results of other **similarly designed studies**.

Discussion/Conclusion

-  Trial may be discussed in comparison to other trials assessing the intervention or the disease state under investigation.
-  Clinical trial **strengths and limitations** should be identified and discussed.
-  The discussion section should also address **future concerns** and **unanswered questions**.
-  The conclusion section should provide an overall research recommendation to the readers.

Discussion/Conclusion

 The investigators' conclusion should focus on the primary endpoint results.

 **Conclusions** should be limited to only that information discussed in previous sections of the trial; no new information should be discussed in this section.